Individual assignment 1 – IoT fundamentals

Coverage: Week 1 -5

Name:

Student ID:

Section A – Multiple choice (20 marks – 10 questions x 2 marks)

1. Which statement best describes a complete IoT system?

A. A sensor that measures temperature

B. A microcontroller running a Python program

C. A system that senses, communicates, processes data, and supports an application

D. Any device connected to Wi-Fi

2. Which device primarily converts a physical quantity into measurable data?

A. Actuator

B. Sensor

C. Router

D. MQTT broker

3. A sensor records one temperature value every 30 seconds.

A. network traffic

B. time-series data

C. an IP address

D. an MQTT topic

4. Which is an example of a sensor-data quality problem?

A. Missing measurement

B. IPv6 address

C. Wi-Fi connection

D. MQTT subscription

5. What is the main purpose of serialization?

A. To increase wireless range

B. To convert structured data into a representation suitable for transmission or storage

C. To select an IP address

D. To increase sensor accuracy

6. A fitness sensor sends a small amount of data to a nearby smartphone and must operate from a small battery. Which connectivity technology is most appropriate?

A. Ethernet

B. BLE

C. Satellite

D. LoRa for every case

7. Which statement about IEEE 802.15.4 and Zigbee is correct?

A. IEEE 802.15.4 is built on Zigbee

B. They are exactly the same technology

C. Zigbee can use IEEE 802.15.4 as its low-power PHY/MAC foundation

D. Zigbee is a transport protocol such as TCP

8. Which transport protocol provides a connection-oriented, reliable, ordered byte stream?

A. IP

B. UDP

C. TCP

D. MQTT

9. Which application protocol is based on publish–subscribe communication?

A. HTTP

B. MQTT

C. IPv6

D. Ethernet

10. An MQTT alarm message is important and should be delivered, but receiving the same alarm twice is acceptable.

A. QoS 0

B. QoS 1

C. QoS 2 is always required

D. MQTT has no delivery levels

Section B – Short Answer (30 marks – 5 questions x 6 marks)

11. IoT Device vs IoT System.

A smartwatch measures a user's heart rate and displays the measurement on its own screen.

Explain the difference between:

1. an IoT device, and
2. a complete IoT system.

Use the smartwatch example in your explanation

12. Sensor data

A room-temperature sensor produces:

10:00	22.5 °C
10:01	22.7 °C
10:02	missing
10:03	22.8 °C
10:04	22.9 °C

Answer:

a. What type of data is this?

b. Identify one data-quality problem.

c. Explain one possible way an application could handle the problem.

13. Data Representation

```
sensor_data = {
    "device": "room01",
    "temperature": 22.8
}
```

And:

```
{"device": "room01", "temperature": 22.8}
```

Explain:

a. What is the first representation?

b. What is the second representation?

c. Why might serialization be required before transmitting the information?

14. TCP and UDP

Explain two differences between TCP and UDP.

Your answer should refer to:

- connection establishment;
- delivery behavior.

Do not simply state that one is "faster."

15. Request–Response vs Publish–Subscribe

Explain the difference between:

1. request–response; and
2. publish–subscribe.

Give one suitable IoT example for each.

Section C – Applied Scenario Questions (40 marks – 2 questions x 20 marks)

16. Smart Greenhouse IoT System

A greenhouse contains battery-powered soil-moisture sensors distributed across a large farm. Each sensor:

- sends a small measurement every 15 minutes;
- may be located several kilometres from the farm office;
- must operate for a long time using battery power;
- does not need to send images or video.

a. Identify three communication requirements that are important when selecting connectivity for these sensors. **(3 marks)**

b. Choose a suitable connectivity technology.

Possible examples include:

- Wi-Fi
- BLE
- Ethernet
- LoRa
- cellular IoT

Give two reasons for your choice. **(5 marks)**

c. The gateway receives:

```
{
  "sensor": "soil07",
  "moisture": 41.2
}
```

What is JSON providing in this system? **(3 marks)**

d. Suppose many greenhouse sensors continuously report measurements to:

- a dashboard;
- a database;
- an irrigation-control application.

Would request–response or publish–subscribe be more suitable?

Explain. **(4 marks)**

e. Name a suitable application protocol for the communication model selected in part (d).

Explain the role of the broker if your selected protocol uses one. **(5 marks)**

17. Campus Environmental Sensor

A university installs battery-powered environmental sensors inside buildings.

Each sensor:

- has limited memory and processing capability;
- sends small temperature and humidity messages;
- uses IEEE 802.15.4;
- should participate in an IPv6-based IoT network;
- is occasionally queried by a local application.

a. Which application protocol is suitable for a constrained request–response device? **(2 marks)**

b. Which transport protocol is commonly used with this application protocol? **(2 marks)**

c. What technology can adapt IPv6 communication to an IEEE 802.15.4 constrained network? **(3 marks)**

d. Complete the protocol stack: **(5 marks)**

Environmental Application

↓

_____

↓

_____

↓

IPv6

↓

_____

↓

IEEE 802.15.4

e. Explain the role of each of the following: **(8 marks)**

- application protocol;
- transport protocol;

Section D — Integrated IoT Communication (10 marks)

18 — Trace the Data

Consider the following IoT system:

Temperature Sensor
↓
Python application
↓
JSON
↓
Wi-Fi
↓
IP
↓
TCP
↓
MQTT
↓
Broker
↓
Monitoring Application

Explain the role of each of the following:

Technology	Role
JSON	
Wi-Fi	
IP	
TCP	
MQTT	